

TA Analyzer - Quick Guide

This document is the **essential operating sequence** limited to the contents of the attached `Manual.pptx` (5 slides). For detailed option explanations, see `Manual_Detailed_EN.md`.

0. Launching the program (Slide 5)

- 1 Start **Anaconda Prompt**.
- 2 Use `cd` to navigate to the project folder (e.g. `cd C:\Users\watqd\OneDrive\KNU_ULSIL\Python_Code\TA_Analyzer_rev5`).
- 3 Run `python ta_main.py` to launch the program.
- 4 Open a measurement file (CSV) with the **Load Data...** or **Load Data (Custom)...** button on the top toolbar.

If the file is in a recognized format, **Load Data...** alone is enough. If your acquisition format differs, use **Load Data (Custom)...** (Section 1).

1. Load Data (Custom) - manual X / Y / Z region selection (Slide 1)

- 1 **Select the X-axis region** - drag-select a single row OR single column that holds the wavelengths
- 2 Click **Confirm This Selection**
- 3 **Select the Y-axis region** - drag-select a single row OR single column that holds the delays
- 4 Click **Confirm This Selection**
- 5 **Select the Z-Matrix region** - drag-select the 2D block of deltaA values (its shape must match the X and Y lengths)
- 6 Click **Confirm This Selection**
- 7 Click **Load**

Tip: while in the table, **Ctrl + Shift + ↑ / ↓ / ← / →** extends the selection to the end of the contiguous data run, just like Excel.

2. Background Correction (Slide 2)

- 1 Click **Background Correction...** on the toolbar
 - 2 Set **# of initial delay points to average** (uses the mean of the first N delays as background)
 - 3 Click **Apply & Close**
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3. Chirp Correction (Slide 3)

- 1 Click **Chirp Correction...** on the toolbar
 - 2 Narrow the visible delay range with **View t_{min}** and **View t_{max}** so the chirp is clearly visible
 - 3 Click the 2D map to add **t points** along the chirp
 - 4 Inspect the dashed fit curve and click **Fit & Apply**
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4. Kinetic Fit - single trace (Slide 4)

- 1 Open **More Analysis** → **Kinetic Fit (single trace)...** on the toolbar
- 2 Set the fit range using **Fit window (delay)** From / To
- 3 Choose the number of **Components** and check each component's **τ_{init}** initial value
- 4 Click **Run Fit**